

1. Overall description of jpeg_Raw image data

The JPEG_Raw image data consists of 4 parts:

Encoded image + file header parameters + raw data

The data is stored in the following formats:

JPEG image
Raw file flags
Raw file header parameters
Raw data

Field descriptions are shown in the table:

Field Name	Offset description		byte count
JPEG image	0	JPEG image data	JPEG image size N
Raw file flag N		Raw file start mark	8
Raw file header parameter	N+8	Description of Raw Image Information 3844	(Fixed Length)
Raw image data N+3852	Raw image data		The raw image is 2 bytes wide x 2 bytes high.

Remark:

Raw file flags

1. The Raw file marker occupies 8 bytes and contains the string "DL700BV", which marks the beginning of the Raw file.

The Raw file header parameters are saved immediately following the Raw file flags.

Raw file header parameters:

1. This field is always 3844 bytes long and describes various information about the Raw image.

Raw image data:

1. This field stores the valid data of the Raw image. Each pixel in the Raw image occupies 2 bytes.

Therefore, a single Raw image occupies 2 bytes (width x height).

2. The width and height information of the raw image is stored in the parameter field of the raw file header.

3. The image data is in little-endian byte order.

2. Query Raw image dimensions and valid data

To read the width and height information of the **Raw** image from the header parameter field of the **Raw** file , follow these steps:

1. View the hexadecimal data of the image file, search for the ASCII code (44 4C 37 30 30 42 56 20) of the string "DL700BV" to find the location of the Raw file marker field.
2. Starting from the 9th byte after the Raw file identifier field, the next 4 bytes are the raw image.

The width is given by the first byte, followed by 4 bytes representing the height of the raw image.

In the image below, the width of the raw image is 0x0280 = 640, and the height is 0x01E0 = 480, both being little-endian bytes.

Save the sequence.

3. Calculate the temperature based on the raw pixel grayscale values.

1. Calculate pixel grayscale values

In raw image data, each 2 bytes represents a pixel, stored in little-endian byte order. View the hexadecimal grayscale.

Degree value, converted to decimal.

2. Grayscale value to temperature value

conversion: Assuming the pixel grayscale value is n

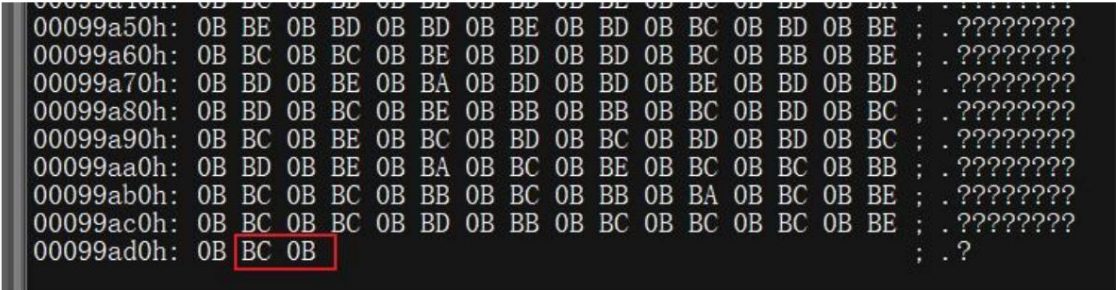
Typical temperature measurement scenario: $temp = (n \times 10 - 27300) / 100$

Extremely high temperature: $temp = (n \times 100 / 4) / 100$

Taking the grayscale value of BC 0B as an example, its decimal equivalent is 0x0BBC = 3004.

In a typical temperature measurement scenario: $temp = (3004 \times 10 - 27300) / 100 = 27.4\ddot{y}$

Extremely high temperature scenario: $temp = (3004 \times 100 / 4) / 100 = 751\ddot{y}$



4. Locate the valid data in the raw section of the JPEG file.

1. Starting from the beginning of the captured JPEG image, locate the position of the valid raw data in a forward direction, step by step.

The steps are as follows:

(1) Locate the **raw** file marker content

The raw file identifier is a fixed string "DL700BV", which, when viewed in hexadecimal, is "44". Finding eight consecutive bytes 4C 37 30 30 42 56 20" , containing this string (data storage order is not affected by the string order)...

(Sectional influence), which refers to the content of the Raw file identifier; the raw file is immediately following this data.

Header parameters.

(2) Locate the valid data location of the **raw** image.

The raw file flags field is always 8 bytes, and the raw file header parameters field is always 3844 bytes. ,
Offset the raw file flag field by 3852 bytes to the beginning, and you'll find the valid data for the raw image.
beginning.

2. Starting from the end of the captured JPEG image, reverse the process to locate the position of the valid raw data (which must already be present).

To find the width and height of the raw image, follow these steps:

(1) Calculate the number of bytes occupied by the raw image.

Raw image data bytes = width x height x 2 (2) The pointer
moves from the end of the JPEG file to the position of valid raw image data.

After reading the JPEG image data, move the pointer to "start from the end of the file and move forward to get the raw image valid data".
The pointer is positioned at the offset of the number of bytes occupied, at which point it is at the beginning of the valid data of the raw image.